

The annual number of forest fires in a certain county in California has Poisson distribution with parameter θ , independently of other years. During three consecutive years, there were 0, 1, and 0 forest fires. Assume an improper non-informative prior $\pi(\theta) = \frac{1}{\theta}$, for $0 < \theta < \infty$.

Is there a significant evidence that θ , the annual frequency of forest fires, does not exceed 1 fire per year? Assume equal losses for Type I and Type II errors.

SOLUTION. We need to test $H_0 : \theta > 1$ vs $H_A : \theta \leq 1$ (because a significant evidence is only needed to reject H_0 in favor of H_A).

For $\mathbf{X} = (X_1, X_2, X_3) = (0, 1, 0) \sim \text{Poisson}(\theta)$,

$$f(\mathbf{X} \mid \theta) = \prod_{i=1}^3 \frac{e^{-\theta} \theta^{X_i}}{X_i!} \sim e^{-\theta} \theta^0 \cdot e^{-\theta} \theta^1 \cdot e^{-\theta} \theta^0 = e^{-3\theta} \theta.$$

Then, the posterior density is

$$\pi(\theta \mid \mathbf{X}) \sim f(\mathbf{X} \mid \theta) \pi(\theta) \sim e^{-3\theta} \theta \cdot \frac{1}{\theta} = e^{-3\theta}.$$

This is **Exponential** density with parameter $\lambda = 3$ and cumulative distribution function $F(\theta \mid \mathbf{X}) = 1 - e^{-3\theta}$. Then,

$$\begin{aligned} \mathbf{P}\{H_A \mid \mathbf{X}\} &= \mathbf{P}\{\theta \leq 1\} = F(1 \mid \mathbf{X}) = 1 - e^{-3} = 0.9502 \\ \mathbf{P}\{H_0 \mid \mathbf{X}\} &= \mathbf{P}\{\theta > 1\} = 1 - F(1 \mid \mathbf{X}) = e^{-3} = 0.0498 \end{aligned}$$

With equal losses, we have a sufficient evidence supporting H_A . So, **yes, there a significant evidence that the annual frequency of forest fires does not exceed 1 fire per year.**

Inventory of some distributions

Discrete families

Poisson(θ)

Generic description:	Number of “rare events” during a fixed time interval
Range of values:	$x = 0, 1, 2, \dots$
Parameter:	$\theta \in (0, \infty)$, frequency of “rare events”
Probability mass function:	$P(x) = \mathbf{P}\{X = x\} = e^{-\theta} \frac{\theta^x}{x!}$
Expectation:	$\mu = \theta$
Variance:	$\sigma^2 = \theta$
Relations:	Limiting case of <i>Binomial</i> (n, p) when $n \rightarrow \infty$, $p \rightarrow 0$, $np \rightarrow \theta$

Continuous families

Exponential(λ)

Generic description:	In a Poisson process, time between consecutive events
Range of values:	$x > 0$
Parameter:	$\lambda \in (0, \infty)$, frequency of events, inverse scale parameter
Density:	$f(x) = \lambda e^{-\lambda x}$
Cdf:	$F(x) = \mathbf{P}\{X \leq x\} = 1 - e^{-\lambda x}$
Expectation:	$\mu = 1/\lambda$
Variance:	$\sigma^2 = 1/\lambda^2$
Relations:	Special case of <i>Gamma</i> (α, λ) when $\alpha = 1$ and <i>chi-square</i> with $\nu = 2$ degrees of freedom. A sum of α independent <i>Exponential</i> (λ) variables is <i>Gamma</i> (α, λ)

Gamma(α, λ)

Generic description:	In a Poisson process, the time of the α -th event
Range of values:	$x > 0$
Parameters:	$\alpha \in (0, \infty)$, shape parameter $\lambda \in (0, \infty)$, frequency of events, inverse scale parameter
Density function:	$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$
Expectation:	$\mu = \alpha/\lambda$
Variance:	$\sigma^2 = \alpha/\lambda^2$
Relations:	For integer α , <i>Gamma</i> (α, λ) is a sum of α independent <i>Exponential</i> (λ) variables $\text{Gamma}(1, \lambda) = \text{Exponential}(\lambda)$ $\text{Gamma}(\nu/2, 1/2) = \text{Chi-square}(\nu)$ As a prior distribution, Gamma family is conjugate to the <i>Poisson</i> (θ) model.